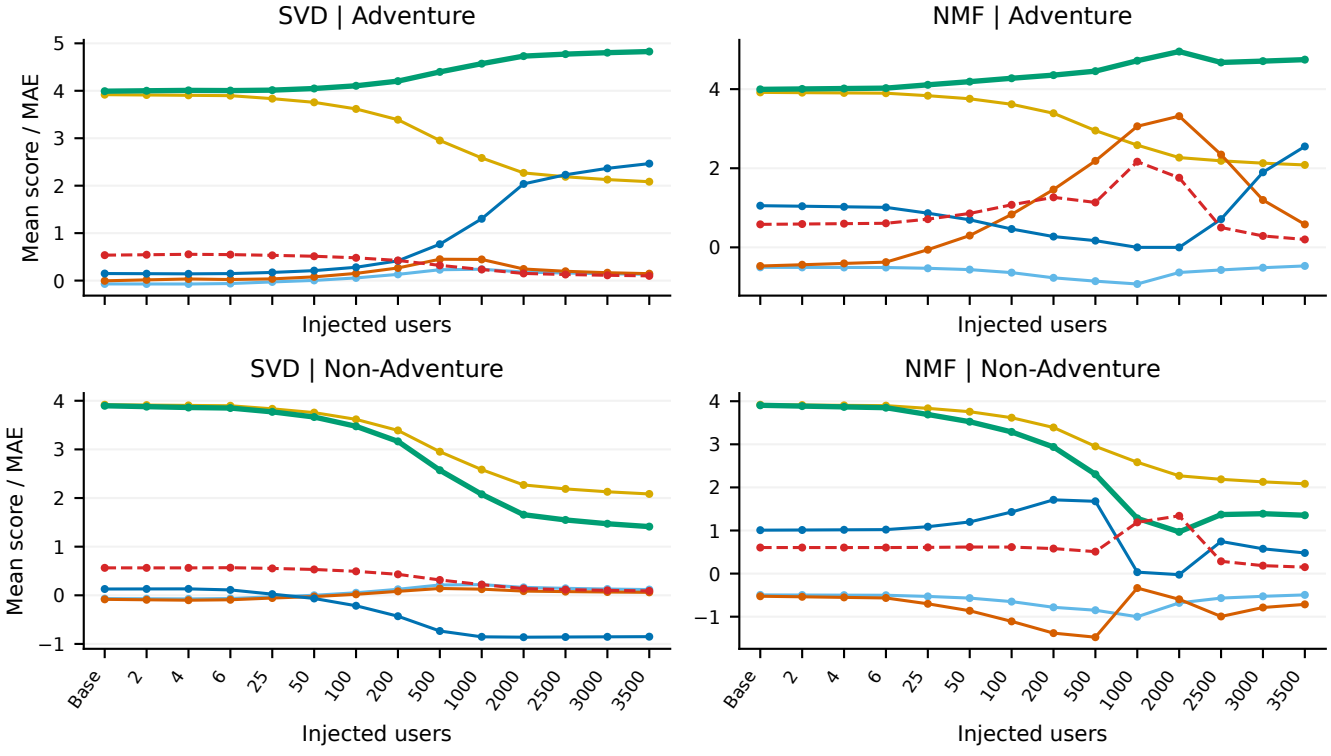
All observed training ratings

$$\hat{r}_{ui} = \mu + b_u + b_i + p_u^\top q_i$$




NMF factor sign check failed at n = 1000, 2000, 2500. Raw results retained.